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Professional Summary

- **Senior Data Engineer** with **10+ years** of experience architecting, developing and optimizing large-scale data pipelines and analytics ecosystems, enabling data-driven decision-making and business growth using modern cloud and big data technologies.
- Recognized expert in modern data platform and lakehouse architecture, with deep expertise in **Databricks**, **Delta Lake**, **Apache Iceberg** and **Apache Hudi** for building scalable, fault-tolerant and ACID-compliant data lakes.
- Strong experience in distributed data processing and cluster optimization across **AWS EMR**, **Databricks** and **Hadoop** environments supporting advanced analytics and machine learning workloads.
- Advanced proficiency in big data processing frameworks and programming languages, including **Apache Spark**, **PySpark**, **Scala**, **Python**, **Java**, **Pandas**, **NumPy** and Hadoop ecosystem tools such as **Hive**, **Presto**, **MapReduce** and **HiveQL**.
- Extensive experience in enterprise ETL and ELT development using **AWS Glue**, **Azure Data Factory**, **Informatica PowerCenter**, **Talend**, **Apache NiFi**, **Dagster**, **IBM DataStage**, **Sqoop** and **Flume**.
- Deep expertise in workflow orchestration and pipeline scheduling using **Apache Airflow**, **Prefect**, **Dagster**, **Luigi**, **AWS Step Functions** and **Azure Logic Apps** to ensure reliable and scalable data operations.
- Proven experience in building real-time and event-driven data architectures using **Apache Kafka**, **AWS Kinesis**, **Apache Flink**, **Azure Event Hubs**, **Azure Stream Analytics**, **SNS**, **SQS** and **Azure Event Grid**.
- Strong background in enterprise data warehousing and analytics platform design using **Snowflake**, **Amazon Redshift** and **Azure Synapse** to deliver high-performance analytical solutions.
- Advanced SQL and database performance optimization specialist with experience across **PostgreSQL**, **SQL Server**, **MySQL**, **Oracle**, **MongoDB** and **AWS RDS**.
- Hands-on expertise in containerization and orchestration using **Docker**, **Kubernetes**, **AWS ECS**, **AWS Fargate**, **AKS** and **OpenShift** for scalable data platform deployments.
- Strong advocate of Infrastructure as Code (IaC), utilizing **Terraform**, **AWS CloudFormation**, **Azure ARM Templates**, **Helm** and **Ansible** for automated and consistent infrastructure provisioning.
- Expertise in CI/CD and DevOps automation using **Jenkins**, **GitHub Actions**, **Azure DevOps**, **AWS CodeBuild** and **SonarQube** to enable efficient and scalable engineering practices.
- Extensive experience in observability and reliability engineering using **Grafana**, **Prometheus**, **ELK Stack**, **Splunk** and **Ganglia** to monitor and optimize data platform performance.
- Experience in end-to-end machine learning platform enablement, including deep learning and neural network model development using **TensorFlow** and **PyTorch**, along with **AWS SageMaker** (Data Wrangler, Pipelines), **Azure ML Studio**, **MLflow** and **Kubeflow**.
- Advanced practitioner in Generative AI and intelligent data solutions, including Large Language Models (LLMs), Agentic AI systems, Retrieval-Augmented Generation (RAG) and vector database integrations, leveraging **OpenAI APIs**, **LangChain** and advanced prompt engineering to build context-aware, AI-driven data applications.
- Strong foundation in data modeling, governance, lineage and metadata management using **Erwin**, **dbt**, **dbForge Studio**, **Lucidchart**, **Azure Purview** and **AWS Glue Data Catalog**.
- Deep understanding of data security, privacy and regulatory compliance (GDPR, HIPAA), implementing secure data practices using **AWS Macie**, **AWS KMS**, **IAM** and **Azure Key Vault**.
- Business intelligence-focused data engineering expertise, delivering actionable insights through **Tableau**, **Power BI**, **Looker** and **Amazon QuickSight**.
- Experience in designing and building scalable data APIs and services using **Flask**, **FastAPI**, **GraphQL**, **Django** and **Django ORM**.
- Strong commitment to engineering excellence and collaboration, utilizing **Git**, **GitHub** and **SVN**, structured branching strategies and peer-review workflows.
- Agile-oriented Senior Data Engineer with proven experience working in cross-functional teams using Scrum methodologies and tools such as **Jira** and **Confluence**.

Professional Experience

Roles & Responsibilities:

- Architected and scaled high-performance ETL/ELT frameworks using **Python**, **Pandas** and **PySpark**, improving transformation efficiency by 40% across multi-terabyte, enterprise-scale datasets.
- Engineered distributed batch data pipelines using **Apache Spark** on **AWS EMR**, reducing end-to-end processing time from 6+ hours to under 2 hours while optimizing cloud compute costs by ~30%.
- Designed and implemented real-time streaming architectures using **AWS Kinesis**, **Apache Kafka** and **Apache Flink**, enabling low-latency ingestion for real-time analytics, ML feature engineering and event-driven AI use cases.
- Developed and deployed high-performance, scalable APIs using **FastAPI** and **Flask**, enabling enterprise-wide data access and AI-driven services with millisecond-level response times.
- Architected and operationalized a **Databricks Delta Lake** lakehouse platform, enabling ACID-compliant data pipelines and high-quality datasets optimized for analytics, ML and GenAI workloads, reducing reconciliation issues by 35%.
- Designed secure, highly available and scalable cloud data platforms using **AWS EC2**, **AWS S3**, **AWS IAM**, **AWS VPC** and **AWS Route Tables**, delivering enterprise-grade, AI-ready Data Lake and analytics ecosystems.
- Developed analytics- and ML-optimized enterprise data models using **Erwin**, **dbForge Studio** and **dbt**, reducing data redundancy by 20% and accelerating downstream analytics and feature engineering.
- Led strategic migration of legacy data warehouses to **AWS Redshift** and **Snowflake**, improving query performance by 55% while significantly reducing infrastructure and licensing costs.
- Built scalable, metadata-driven ingestion frameworks using **AWS Glue Crawlers**, **AWS Glue Data Catalog** and **AWS Glue Pipelines**, enabling automated schema evolution and reducing onboarding time for 100+ data sources by 60%.
- Engineered production-grade ML platforms using **AWS SageMaker** and **Kubeflow**, enabling scalable training and deployment of neural network based models, with automated retraining pipelines to mitigate model drift.
- Developed and deployed **LLM**-powered and **Agentic AI** driven data products using **OpenAI APIs**, large language models (**LLMs**) and advanced prompt engineering, enabling automated SQL generation, intelligent workflow orchestration and conversational analytics, improving productivity by 40%.
- Designed and implemented robust **RAG** (Retrieval-Augmented Generation) architectures integrating **LLMs** with enterprise data sources to deliver context-aware, explainable and accurate AI-driven insights.
- Built and optimized **vector**-based retrieval systems using **vector databases**, enabling semantic search, similarity matching and efficient information retrieval for **LLM** and **RAG**-powered applications.
- Designed cost-efficient, serverless analytics solutions using **AWS Athena**, reducing BI query costs by ~70% while enabling scalable, on-demand data exploration.
- Modernized data and ML workloads through containerization using **Docker**, **AWS ECS** and **Kubernetes**, improving scalability, deployment consistency and reducing infrastructure overhead by 30%.
- Unified and modernized ingestion ecosystems using **Informatica PowerCenter** and **Apache Airflow**, transitioning legacy batch pipelines to near real-time, orchestrated workflows.
- Engineered highly available, resilient data services using **SQLAlchemy** with **AWS RDS (PostgreSQL and SQL Server)**, ensuring reliable transactional processing and seamless system integration.
- Developed internal enterprise-grade applications using **Django ORM**, automating reporting workflows and reducing manual effort by 50% for finance and compliance teams.
- Implemented enterprise-grade data governance, security and privacy controls using **AWS Macie**, **AWS KMS** and **GDPR** compliance frameworks, ensuring audit readiness and protection of sensitive data assets.
- Established comprehensive observability and monitoring frameworks using **Grafana** and **Prometheus**, enabling proactive alerting and reducing production incidents by 40%.
- Orchestrated complex data, ML and GenAI workflows using **AWS Step Functions** and **Luigi**, enhancing pipeline reliability, observability and SLA adherence across distributed systems.
- Standardized infrastructure provisioning using **Terraform** and **AWS CloudFormation**, enabling fully automated, reusable and version-controlled environment management.
- Streamlined and optimized CI/CD pipelines using **Jenkins** and **AWS CodeBuild**, increasing deployment frequency from bi-weekly to daily while improving release stability and automation.
- Championed collaborative software engineering best practices using **GitHub**, enhancing code quality, peer review efficiency and cross-functional team collaboration.

- Delivered executive-level, self-service dashboards using **Tableau**, enabling real-time, data-driven decision-making and reducing reporting turnaround from weeks to hours.
- Led **Agile/Scrum** delivery processes, consistently achieving a 95% sprint success rate while aligning engineering execution with evolving business and AI-driven priorities.

Client: Centene Corporation ,St. Louis ,MO
Senior Data Engineer

Dec 2022 - Jul 2024

Roles & Responsibilities:

- Architected and delivered enterprise-scale data platforms leveraging **Apache Spark**, **Azure Databricks** and **Delta Lake**, enabling high availability, fault tolerance and performance-optimized analytics at scale.
- Led the design and evolution of distributed ETL frameworks using **PySpark**, **Scala** and **Python**, boosting data processing throughput by 45% for multi-terabyte workloads.
- Engineered and automated end-to-end data ingestion and transformation pipelines using **Azure Data Factory** and **Informatica**, reducing execution time by 50% while enforcing robust data quality and validation standards.
- Designed and deployed real-time streaming architectures using **Azure Event Hubs**, **Apache Kafka** and **Apache Flink**, powering low-latency analytics and fraud detection systems.
- Established scalable orchestration and workflow automation frameworks using **Apache Airflow**, **Azure Logic Apps** and **Prefect**, ensuring SLA compliance and resilient pipeline execution across multi-cloud environments.
- Designed and implemented secure, highly available cloud infrastructure using **Azure Virtual Machines**, **Azure Blob Storage**, **Azure Route Tables**, **Azure Active Directory** and **Azure Virtual Networks**, consistently achieving 99.9% uptime.
- Championed Infrastructure-as-Code (IaC) and automation strategies using **Terraform**, **ARM Templates**, **Ansible** and **Helm**, reducing manual deployments by 80% and strengthening DevSecOps governance.
- Built and optimized CI/CD pipelines using **Azure DevOps** and **Jenkins**, accelerating release cycles by 35% and ensuring seamless, zero-downtime deployments.
- Implemented enterprise-grade observability and monitoring frameworks using **ELK Stack** and **Splunk**, improving incident response times by 40% and enhancing operational resilience.
- Designed and optimized modern data warehouse solutions using **Azure Synapse Analytics** and **Snowflake**, improving query performance by 60% and enabling scalable self-service analytics.
- Optimized large-scale analytical workloads across **Azure Data Lake Analytics**, **Databricks SQL** and **Apache Hive**, reducing processing costs by 30% while supporting petabyte-scale data volumes.
- Standardized enterprise data modeling, transformation and documentation practices using **Erwin**, **dbt** and **Lucidchart**, ensuring data lineage visibility, modularity and cross-team consistency.
- Integrated and fine-tuned **PostgreSQL** systems to support hybrid OLTP and OLAP workloads, enabling high-performance and scalable enterprise analytics.
- Established robust data governance, security and compliance frameworks using **Azure Purview** and **Azure Key Vault**, ensuring GDPR compliance and strengthening enterprise data protection.
- Built and operationalized end-to-end machine learning pipelines using **Azure Machine Learning** and **MLflow**, significantly reducing model deployment cycles from weeks to days.
- Drove AI innovation through **prompt engineering**, **LangChain** and **OpenAI APIs**, enabling intelligent automation, advanced summarization and conversational analytics solutions.
- Delivered executive-level dashboards and reporting solutions using **Power BI**, reducing time-to-insight by 50% and enabling data-driven strategic decision-making.
- Designed and developed scalable APIs and applications using **Flask**, **FastAPI**, **GraphQL** and **Django**, while leading Agile delivery using **Jira** and **Confluence** to ensure aligned, on-time execution of enterprise initiatives.

Client: Comerica, Dallas, TX
Data Engineer

Feb 2020 - Nov 2022

Roles & Responsibilities:

- Spearheaded the architecture and evolution of enterprise-scale data platforms on **Azure Synapse Analytics**, integrating **Azure Data Lake Storage** and **PolyBase** to deliver high-performance, scalable analytical ecosystems.
- Architected and optimized distributed data processing frameworks using **PySpark** on **Databricks**, orchestrating petabyte-scale ETL pipelines and increasing processing throughput by 40%.
- Delivered high-efficiency data transformation solutions leveraging **Databricks**, **Apache Spark** and **Delta Lake**, reducing pipeline runtime from 8 hours to 1 hour while significantly improving data freshness and reliability.

- Designed and implemented real-time streaming pipelines using **Azure Stream Analytics** and **Kafka**, enabling fraud detection and anomaly detection with sub-3-second latency.
- Engineered scalable, event-driven architectures using **Azure Functions** and **Event Grid**, streamlining real-time integrations and reducing infrastructure costs by 25%.
- Directed ingestion and integration of complex, heterogeneous data sources using **PyODBC**, automating pipelines from on-premise into Synapse with 99.9% reliability and resilient error-handling mechanisms.
- Led the development of reusable, enterprise-grade ETL frameworks in **Azure Data Factory**, implementing parameterized, secure and auditable Data Flows to standardize data transformation processes.
- Designed and governed workflow orchestration using **Dagster** and **Apache Airflow**, ensuring fault-tolerant execution, dependency management and high availability across mission-critical pipelines.
- Developed and deployed scalable **RESTful APIs** using **Flask** and **FastAPI**, enabling seamless, real-time access to curated datasets and machine learning outputs.
- Operationalized end-to-end machine learning pipelines using **Azure ML Studio** and **MLflow**, enabling reproducible experimentation, CI/CD integration and improving forecasting accuracy by 20%.
- Enabled enterprise-wide self-service analytics by integrating curated data assets into **Power BI** and **Tableau**, reducing reporting cycles from weeks to hours and enhancing decision-making agility.
- Optimized enterprise **Oracle** databases through advanced **SQL** tuning, partitioning and indexing strategies, achieving a 30% reduction in query latency and improved system performance.
- Containerized and deployed scalable data services using **Docker** and **Kubernetes (AKS)**, ensuring auto-scaling, high availability and 99.95% uptime in production environments.
- Automated cloud infrastructure provisioning using **Terraform** and **Azure Resource Manager (ARM)** templates, reducing environment setup time by 80% while enforcing Infrastructure-as-Code governance.
- Implemented robust CI/CD pipelines using **Jenkins**, automating testing, version control and deployments to achieve zero-downtime production releases.
- Administered and optimized **Azure VMs**, **Blob Storage** and **Key Vault**, ensuring security compliance, cost efficiency and operational resilience, while leading **Agile** delivery using **Jira**, **Git** and **Confluence** to drive engineering excellence.

Client: Etsy, Inc, Brooklyn,NY
Data Engineer

Dec 2017 - Jan 2020

Roles & Responsibilities:

- Led enterprise-wide data integration using **Apache NiFi**, designing scalable dataflows, flowfile transformations and custom processors to ingest data from APIs, RDBMS and flat files into **AWS S3** and **Amazon Redshift**.
- Built and optimized automated ETL frameworks using **AWS Glue Crawlers** and **PySpark Glue Jobs**, enabling schema evolution, eliminating manual updates and supporting self-service data discovery.
- Developed modular and reusable data pipelines in **Python** leveraging **Pandas** and **NumPy**, accelerating ingestion development and improving engineering productivity across multiple data domains.
- Architected event-driven data pipelines using **AWS SNS**, **AWS SQS**, **AWS Step Functions** and **AWS Lambda** to enable near real-time micro-batch processing, reducing data latency from hours to minutes.
- Engineered scalable streaming data solutions with **Apache Kafka**, implementing topic partitioning, consumer group optimization and schema validation using **Confluent Schema Registry**.
- Orchestrated end-to-end batch and streaming workflows using **Apache Airflow**, developing custom operators, enforcing SLA monitoring and improving pipeline reliability while reducing job failures.
- Utilized **Databricks on AWS** with **Delta Lake** and optimized **Apache Spark** transformations to support large-scale data processing, change data capture (CDC) and feature engineering for analytics and machine learning.
- Implemented Infrastructure-as-Code (IaC) using **AWS CloudFormation** and **Terraform** to provision and manage cloud resources including **AWS S3**, **AWS EMR**, **AWS IAM** and **AWS VPC**, ensuring scalability and eliminating configuration drift.
- Administered and optimized **AWS RDS** environments (**PostgreSQL** and **SQL Server**) using parameter tuning, read replicas and enhanced monitoring to maintain high availability and performance across production systems.
- Containerized and deployed data processing workloads using **Docker** on **AWS Fargate**, improving fault tolerance, scalability and reducing infrastructure costs significantly.
- Designed and maintained CI/CD pipelines using **Jenkins**, integrating **Git**, **SonarQube** and **Docker** to automate build, testing and deployment of data pipelines with minimal errors.
- Developed secure **RESTful APIs** using **Flask** and built internal applications and administrative dashboards using **Django** to enable data access, metadata management and self-service capabilities for analysts and stakeholders.

- Designed and delivered business intelligence solutions using **AWS QuickSight**, integrating **Amazon Athena** and **AWS RDS** to create executive dashboards visualizing key KPIs such as churn, revenue and growth.
- Built and deployed machine learning pipelines using **AWS SageMaker** (including **SageMaker Data Wrangler** and **SageMaker Pipelines**) and optimized **SQL Server** performance through advanced query tuning, indexing and stored procedures to enhance reporting efficiency.

Client: GE HealthCare Technologies Inc, Chicago, IL
ETL Developer

Oct 2015 - Nov 2017

Roles & Responsibilities:

- Architected and delivered end-to-end ETL solutions using **Informatica PowerCenter**, enabling seamless integration of data from diverse source systems into enterprise data warehouses and data marts.
- Engineered highly efficient mappings, sessions and workflows leveraging reusable transformations in **Informatica PowerCenter**, resulting in reduction in ETL processing time.
- Designed robust data extraction frameworks for high-volume datasets from sources including **Oracle**, **SQL Server** and flat files, ensuring data consistency and integrity.
- Developed scalable batch ETL processes using **IBM DataStage**, supporting enterprise-wide data integration and improving system reliability.
- Built and optimized complex SQL scripts, stored procedures and database objects in **Oracle** and **SQL Server**, significantly enhancing transformation logic performance and reducing query execution time.
- Implemented comprehensive data quality frameworks using **Informatica PowerCenter**, including validation rules, deduplication and standardization processes, improving data accuracy and reducing discrepancies by up to 35%.
- Proactively monitored and tuned ETL jobs within **Informatica PowerCenter** and **IBM DataStage**, identifying bottlenecks and optimizing workflows to improve system throughput and runtime efficiency.
- Automated ETL operations using **Unix Shell Scripting**, including scheduling, file handling and error recovery mechanisms, reducing manual intervention and operational overhead.
- Designed and deployed enterprise-grade ETL solutions using **SQL Server Integration Services (SSIS)** for large-scale data migration and system consolidation initiatives.
- Led data validation, reconciliation and testing efforts (Unit Testing, SIT, UAT) across **Oracle** and **SQL Server** environments, ensuring high-quality data delivery aligned with business expectations.
- Partnered with business stakeholders, data architects and cross-functional teams while working with **Informatica PowerCenter** and **IBM DataStage** to translate complex business requirements into scalable and maintainable ETL solutions.
- Produced detailed technical and functional documentation for processes implemented in **Informatica PowerCenter**, **IBM DataStage** and **SSIS**, facilitating audits, support and knowledge transfer.

Certifications

- AWS Certified: Data Engineer Associate
- Microsoft Certified: Azure Data Engineer Associate
- AWS Certified AI Practitioner Foundational

Education

- Bachelors in Computer Science.